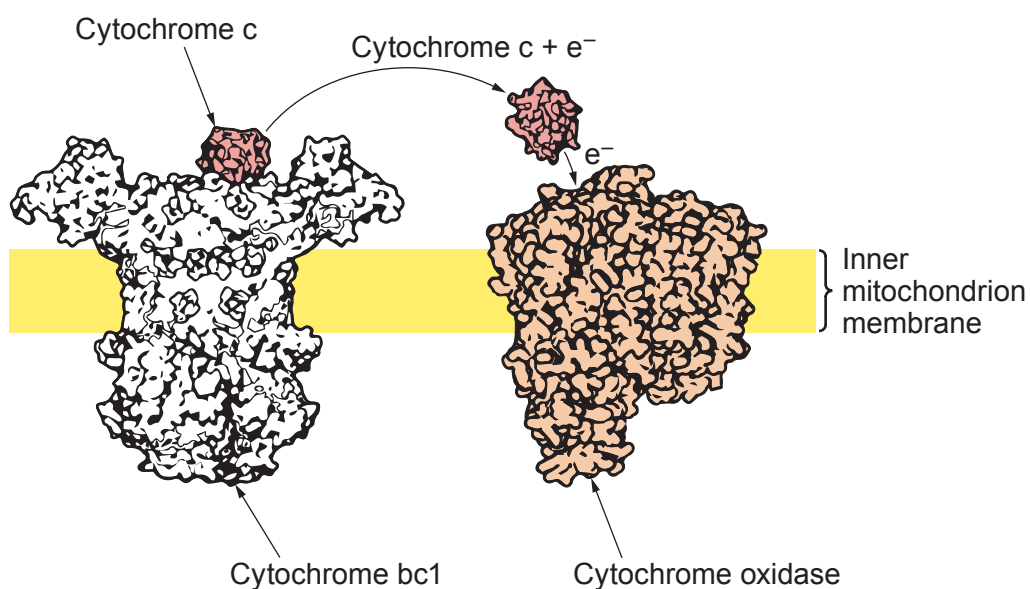


6. Cytochrome c is an electron carrier involved in oxidative phosphorylation. It is a protein that transfers electrons between two large proteins found in the electron transport chain. It removes electrons from cytochrome bc1 and transfers them to cytochrome oxidase. This is shown on the diagram below.



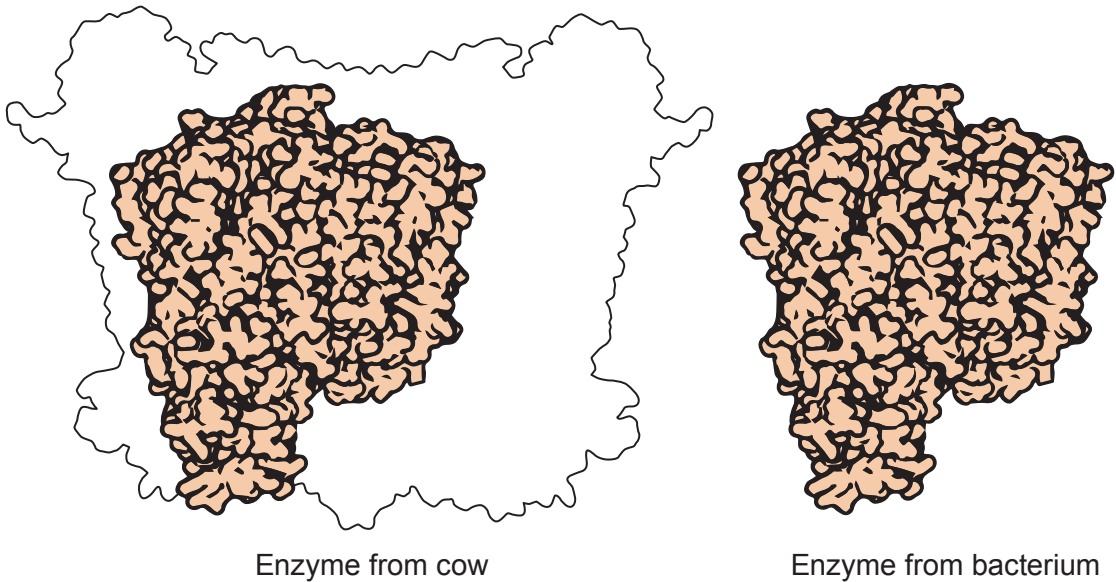
- (a) Cytochrome oxidase is the final protein in the electron transport chain. It is an enzyme which transfers the electrons to an oxygen molecule.

State the final product of this reaction in the mitochondrion.

[1]



(b) The cytochrome oxidase molecule is different in different species. The diagram below shows the enzyme from a cow and a bacterium. The shaded section in each represents the polypeptide chains which are very similar in both organisms. This is the area of the enzyme which binds to the cytochrome c.



(i) Suggest why the shaded regions are similar in all organisms. [2]

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(ii) Suggest why cytochrome c detaches from the cytochrome oxidase molecule when the electrons it carries are removed. [2]

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- (c) Cytochrome bc1 is a proton pump that transfers protons into the inter-membrane space. Antimycin is a chemical that binds to cytochrome bc1 and blocks its action.

Explain what happens to the pH of the intermembrane space when antimycin is present. [2]

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- (d) Oligomycin A is another chemical which acts by inhibiting protons from passing through ATP synthase.

- (i) Explain why the presence of oligomycin A in a human causes high levels of reduced NAD to form in the cell. [2]

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- (ii) Suggest why lactate begins to accumulate in the blood if high levels of reduced NAD form. [2]

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